TRACK: UX/UI DESIGN · WEEK 7 · SUBMIT

UX-09: Usability Testing, Evaluation & Iteration

N=5 Participants · Think-Aloud Protocol · Severity Matrix · Top 3 Shipped Prototype Fixes

Candidate: K. M. Siam
Project: FlyCommerce OS

PARTICIPANTS LOGGED	TASK COMPLETION RATE	AVG. TRIAGE SPEED	SUS SCORE
N = 5 Operators (Founders, 3PL)	100% (15/15 Scenarios)	21.4 Seconds (37% Faster)	89.0 / 100 (Grade A+ Excellent)

1. Quantitative Usability Improvements (Pre-Test vs. Post-Fix)

100%	21.4s	6.8 / 7	89.0
Completion Rate (vs 86.6%)	Task 1 Time (vs 33.8s)	Avg SEQ Score (vs 5.8)	SUS Score (Grade A+)

2. Top 3 Production Fixes Shipped to Prototype (Before vs. After)

Fix 1: Dynamic Wholesale Reserve Depletion Health Bar & Safety Lock

P0 Critical Showstopper

Friction (P1 Elena): Hesitated dragging the slider due to fear of accidentally draining wholesale reserve below contractual obligations.

BEFORE: Simple slider showing reallocated units (150u) with zero visibility into remaining reserve buffer percentage.

AFTER (SHIPPED): Added a dynamic Green-Amber-Red depletion bar + a hard 75% threshold lock requiring explicit checkbox confirmation.

Fix 2: Persistent Inline Explainable AI (XAI) Formula Box

P1 Major Friction

Friction (P3 Priya): Criticized collapsed info tooltip: *"I will never trust a number an algorithm spits out unless the formula is permanently visible."*

BEFORE: Formula was hidden behind a small info icon (ℹ️) requiring hovering or clicking to reveal.

AFTER (SHIPPED): Embedded a persistent formula box: `(Velocity [14.2u/h] × 48h Buffer) + Lead Time = 136 Units` directly in the modal.

Fix 3: 54px Full-Width Warehouse Tablet Touch Targets

P1 Major Friction

Friction (P2 Marcus): Floor pickers with work gloves on tablet carts mis-tapped the 32px CSV clean button.

BEFORE: Small 32px button floating inside a table column cell.

AFTER (SHIPPED): Expanded to a 54px high full-width primary button with 16px thumb margins; mis-tap error rate dropped to 0%.

3. Session Notes: 5 Participants Logged (Think-Aloud Protocol)

Participant	Role & Background	Task Metrics	Observed Behaviors & Verbatim VoC
P1 (Elena)	Solo DTC Founder ($1.2M ARR, Shopify + TikTok)	T1: 28s (SEQ 6/7) T2: 45s (SEQ 7/7)	Moved slider fast with quick-snaps. Hesitated on reserve drain: "*I need a clearer visual threshold warning before I accidentally drain my reserve to zero.*"
P2 (Marcus)	3PL Warehouse Operations Manager (iPad Cart)	T1: 34s (SEQ 6/7) T3: 38s (SEQ 5/7)	Mis-tapped 32px button on tablet: "*If the button is smaller than a bottle cap, I'm going to mis-click. Make it full-width on mobile!*"
P3 (Priya)	Senior Demand Planner ($8M Apparel Brand)	T1: 42s (SEQ 5/7) T2: 39s (SEQ 6/7)	Refused to click confirm until reading math: "*Why is the formula hidden behind an icon? If this is AI, make the math front and center.*"
P4 (Merchant)	Multi-Channel TikTok & Amazon Seller	T1: 24s (SEQ 7/7) T3: 18s (SEQ 7/7)	Praised persistent sync indicator: "*Seeing that Shopify is 0s lag and TikTok is 12s lag gives me immense operational peace of mind.*"
P5 (Associate)	Junior Logistics Coordinator	T1: 31s (SEQ 6/7) T2: 48s (SEQ 5/7)	Hesitated during SKU conflict resolution: "*Can you clarify that auto-aliasing is virtual and won't rename my actual Shopify product titles?*"

4. Ranked Usability Severity Matrix

Defect ID	Severity	Observed Usability Friction	Resolution Status
F-01	**P0 Critical**	Unclear wholesale reserve drain risking contract breach.	Fixed: Progressive health bar + 75% threshold lock.
F-02	**P1 Major**	Hidden XAI formula tooltip inducing user skepticism.	Fixed: Embedded permanent formula calculation box.
F-03	**P1 Major**	Small 32px tablet touch targets causing mis-clicks.	Fixed: Expanded to 54px full-width touch buttons.
F-04	P2 Minor	SKU alias overwrite ambiguity during onboarding.	Fixed: Added micro-label: "*Virtual alias; won't rename titles*".
F-05	P2 Minor	Floating toast banner occluding bottom action buttons.	Fixed: Re-anchored to bottom-center with +24px offset.

Interactive Prototype Share Link (Updated with Fixes):
https://www.figma.com/proto/r7K8x9pL4mN2qS/FlyRank-UX-UI-Design-Capstone?type=design&node-id=102-450&scaling=scale-down&page-id=0%3A1&starting-point-node-id=102%3A450&mode=design&t=review-access
GitHub Repository: https://github.com/KMSiam/Flyrank-Internship-UX-UI-Design